

Does Discrete Ricci Curvature Improve Failure-Aware Traffic Engineering? A Controlled Comparison with Direct Scenario-Risk Optimization

Emre Öztürk

*Department of Management Information Systems, Hatay Mustafa Kemal
University, Hatay, Türkiye*

Abstract

Discrete Ricci curvature describes structural fragility, but diagnostic association does not establish routing value. We test the incremental prescriptive value of negative Ollivier–Ricci curvature against capacity-aware and outcome-aligned traffic engineering (TE). Path-based multi-commodity flow on directed REPETITA backbones preserves capacities and origin–destination commodities. All policies share candidate paths and matched maximum-link-utilization (MLU) and latency budgets; one traffic matrix tunes regularization and four remain held out. Resilience is independent frozen-route service loss after physical-link failures, with adaptive capacity-feasible delivery as a recovery ceiling.

On a frozen ten-topology set, curvature did not improve mean single-link-failure delivery: relative to minimum-MLU TE, loss increased by 0.452 percentage points (topology-bootstrap 95% CI: 0.077 to 0.935; exact sign-flip $p = 0.0625$) and path cost by 1.36%. A direct single-link minimax control on ten disjoint topologies reduced worst loss relative to curvature by 2.246 points (95% CI: 0.871 to 3.596 points; $p = 0.03125$). We generalize direct control to expected-loss, CVaR, and minimax programs for multi-link scenarios. On twelve topologies with design/evaluation link-pair splits, CVaR(0.90) TE reduced held-out CVaR loss relative to curvature by 1.297 points (95% CI: 0.554 to 2.074 points; $p = 0.015625$), while mean loss increased by 0.281 points and path cost by 1.90%. Curvature conclusions survived path-count,

Email address: `emre.ozturk@mku.edu.tr` (Emre Öztürk)

MLU-slack, and laziness checks; tail gains weakened with only 40 design scenarios. Geometry is useful diagnostically, but direct risk objectives better control the declared failure-loss tail and make its mean–tail trade-off explicit.

Keywords: traffic engineering, network resilience, multi-commodity flow, Ollivier–Ricci curvature, Forman–Ricci curvature, link failure, REPETITA

1. Introduction

Traffic engineering (TE) determines how a traffic matrix is carried by a capacitated network. Classical objectives balance load, minimize delay, or protect the network against demand and topology changes [1, 2]. Link failures make this problem explicitly structural: a routing that is satisfactory in the nominal topology may lose a large fraction of traffic before a controller converges or installs a new configuration. Reproducible frameworks such as REPETITA therefore evaluate both stored configurations under failure and reoptimization after failure [3].

That literature has moved through increasingly explicit treatments of uncertainty. Nominal link-weight optimization first centered on utilization and delay; robust and oblivious formulations then protected against uncertain demand sets [4, 5]. Failure-aware systems added precomputed or rapidly installed alternatives [6, 2], while TeaVaR introduced distributional availability risk into WAN TE [7]. Recent DOTE, outcome-aware mitigation, and maximum-load verification evaluate end-to-end behavior under demand and topology changes rather than relying on local surrogates [8, 9, 10]. This trajectory makes the comparison standard for a new routing signal demanding: it should be tested not only against nominal TE, but also against controls that directly encode the declared failure outcome. Discrete curvature enters this history from a different direction, as a geometric descriptor rather than an operational objective.

Discrete Ricci curvature formalizes that geometric view of network structure. Ollivier–Ricci curvature compares probability measures around adjacent nodes through optimal transport [11]; Forman–Ricci curvature provides a cheaper combinatorial alternative [12, 13]. On Internet topologies, curvature is associated with local heterogeneity, betweenness, and connectivity [14]. Across several domains it has consequently been used as a proxy for robustness or vulnerability. These uses are diagnostic: they ask whether a score identifies an edge with a particular structural role. They do not by

themselves show that placing the score in a routing objective improves a network-level outcome.

This distinction matters because a curvature penalty mechanically moves traffic away from negatively curved edges. Measuring success as “less traffic on negative-curvature edges” then reuses the treatment variable as the outcome. Such a comparison cannot tell whether more traffic is delivered when a link actually fails, whether the same effect is available from degree or betweenness, or whether ordinary capacity-aware TE already performs better.

We address the prescriptive question directly:

At matched utilization and latency budgets, does a discrete-curvature regularizer improve service delivery after physical link failures beyond minimum-MLU traffic engineering and curvature-blind structural controls?

The study makes five contributions.

1. It formulates the comparison as path-based multi-commodity TE, retaining OD identities, directed capacities, and a method-neutral candidate-path set.
2. It defines resilience through independent failure outcomes: frozen-route delivered traffic and capacity-feasible delivery on the residual topology.
3. It separates development from confirmation. One traffic matrix tunes regularization; four matrices per topology remain held out, and the primary effect is tested with topology-level rather than link-level inference.
4. It generalizes a single-link exposure control to a scenario-risk LP for expected loss, CVaR, and minimax loss under multi-link failures. The formulation is an outcome-aligned control—not a claim that risk-aware TE is new.
5. It uses disjoint development, single-failure confirmation, tail validation, and topology-plus-scenario double-failure sets. One-factor sensitivity of both curvature and scenario-risk choices, an adaptive recovery ceiling, and a fixed 30–69-node timing sequence probe robustness and cost.

The confirmatory result is negative but informative: Ollivier–Ricci regularization does not improve average single-link-failure delivery. The observed

tail-risk signal is also not unique to geometry; separately held-out validation shows that directly optimizing the relevant failure outcome is more effective. These findings delimit where curvature can be useful: as a structural descriptor or hypothesis generator, not as a substitute for a validated TE objective.

2. Background and related work

2.1. Capacity-aware and failure-aware traffic engineering

Load-balancing TE commonly minimizes maximum link utilization or optimizes link weights against a projected traffic matrix [1]. Robust and oblivious formulations already address uncertain demands: early intradomain work characterized the trade-off between performance on a known matrix and robustness to a demand set, while COPE optimized for dynamic environments [4, 5]. Centralized systems such as B4 and SWAN show how multi-commodity optimization, utilization, and operational path constraints interact in wide-area networks [15, 16].

Failure-aware systems reserve alternatives, precompute forwarding state, or reoptimize after topology changes. R3, forward fault correction, fast segment routing optimization, and semi-oblivious TE provide distinct points in this design space [6, 2, 17, 18]. More recent restoration-aware and black-hole-resilient systems further demonstrate that the recovery mechanism and its failure model must be explicit [19, 20]. The methodological implication is that failure performance must be measured on declared failed topologies or traces, not inferred from a nominal structural score.

Recent systems sharpen that requirement. DOTE evaluates prediction-free WAN TE under demand variation and random multi-link faults, while performance-aware mitigation ranks actions by estimated end-to-end flow outcomes rather than local or global proxy scores [8, 9]. Velo shows that worst link loads can change materially when route changes accompany failures and develops scalable verification for the joint state space [10]. These systems address different control planes, but all reinforce the distinction between a structural proxy and the operational quantity ultimately optimized or verified.

REPETITA was designed to make such comparisons repeatable. It provides real topologies augmented with bandwidth, delay, and traffic matrices, implements multi-commodity-flow lower bounds and established TE algorithms, and includes single-link-failure scenarios with and without reopti-

mization [3]. The underlying real-network corpus is derived from the Internet Topology Zoo [21]; related survivable-design instances are available in SNDlib [22]. Robust validation research also warns against evaluating a design on the same uncertainty set used to construct it [23]. We therefore use REPETITA’s data model and failure distinction while implementing a common path-based optimization layer for all policies. MetaOpt’s adversarial analysis of production TE heuristics provides a complementary lesson: the performance gap between a heuristic and its outcome-aligned comparator is itself an important object of study [24].

2.2. Risk-aware traffic engineering

Probabilistic failure objectives are established in TE. TeaVaR, in particular, uses a Value-at-Risk view to trade utilization against availability under a failure distribution [7]. Expected loss, conditional value-at-risk (CVaR), and worst-case loss are therefore not novelty claims of this paper; the scenario-based CVaR representation follows the standard Rockafellar–Uryasev construction [25]. We use these risk functionals as strong, outcome-aligned controls. The unresolved question is narrower: after capacity, latency, candidate paths, and failure information are controlled, does a topology-only geometric score retain incremental prescriptive value? To prevent direct controls from being tested on their own design scenarios, multi-link scenarios are deterministically split before optimization and only the disjoint subset is used for evaluation.

2.3. Discrete curvature as a structural score

For a graph with metric d and an α -lazy probability measure m_u around node u , Ollivier–Ricci curvature of edge (u, v) is

$$\kappa_{uv} = 1 - \frac{W_1(m_u, m_v)}{d(u, v)}, \quad (1)$$

where W_1 is the Earth Mover’s (1-Wasserstein) distance [11]. Negative values indicate that the two local measures are farther apart than their base nodes and frequently occur at sparse connections between neighborhoods. Internet measurements show relationships with degree, betweenness, connectivity, and geography [14]. These relationships motivate—but do not validate—a routing penalty.

Forman–Ricci curvature is defined from cell-complex incidence and edge/node weights [12]. Its graph specialization is inexpensive, and empirical comparisons show that it overlaps with but is not equivalent to Ollivier curvature [26, 13]. Extensions also treat direction and other complex-network geometry [27]. We therefore treat Forman curvature as a separate baseline rather than an exact surrogate for Ollivier curvature. Curvature-guided hierarchical partitioning and routing has also appeared for heterogeneous communication networks [28]; its goal is scalable decomposition, whereas our target is service loss under capacity-constrained physical failures.

2.4. Gap addressed by this study

Three questions are often conflated: (i) does a score correlate with a structural property, (ii) does it predict the outcome of an actual failure, and (iii) does optimizing that score improve routing? Our design evaluates all routing policies on (iii) using outcomes that do not contain the policy’s own score. It also includes curvature-blind controls to determine whether an effect is geometric or merely a consequence of regularization.

Table 1 makes the novelty boundary explicit. Risk-aware TE already optimizes failure distributions, recent systems optimize or verify end-to-end failure outcomes, and curvature already supports structural analysis and hierarchical routing. To our knowledge, no prior study joins these lines in a matched incremental-value test: identical candidate paths, MLU and latency budgets, independent physical-failure outcomes, curvature-blind controls, and direct scenario-risk optimizers are evaluated on held-out topologies and failure pairs. The contribution is this controlled prescriptive audit and its evidence, not a claim that CVaR, minimax optimization, or curvature is new.

3. Model

3.1. Directed network and physical links

Let $G = (V, A)$ be a directed routing graph. Arc $a \in A$ has capacity $c_a > 0$ and delay $\ell_a > 0$. Reciprocal arcs belong to the same physical link $e \in \mathcal{E}_{\text{phys}}$. A physical-link failure removes both directions. REPETITA parallel arcs with identical delay and IGP weight are represented as one capacity bundle; unequal-cost parallel arcs are excluded rather than approximated.

Let K denote the OD commodities. Commodity k has source s_k , destination t_k , and demand d_k . We generate the same $q = 8$ delay-shortest

Table 1: Closest methodological neighbors and the remaining comparison gap.

Line of work	Failure or uncertainty target	Validation and scale emphasis	Boundary relative to this study
TeaVaR [7]	Probabilistic availability–utilization trade-off using a VaR formulation	WAN failure distributions and risk-aware TE	Establishes that probabilistic risk objectives are not new; does not test the incremental value of a topology-only geometric regularizer.
DOTe [8]	Demand uncertainty with random multi-link faults	Stochastic optimization, learned scaling, and large-WAN evaluation	Addresses prediction-free TE quality and failure recovery; not a matched geometric-score audit.
MetaOpt [24]	Adversarial inputs that expose heuristic performance gaps	Solver-based analysis of production TE and other heuristics	Motivates controlled incremental-value comparisons, but does not study curvature or held-out physical-failure loss.
Outcome-aware failure analysis [9, 10]	End-to-end mitigation quality and worst link loads under failures or route changes	Production incidents, large data centers, and ISP-scale verification	Optimizes or verifies operational outcomes directly; it does not compare those outcomes with curvature-aware routing under common budgets.
Curvature-guided networking [14, 28]	Internet structure and curvature-guided hierarchical partitioning/routing	Structural association and scalable community decomposition	Supplies the geometric motivation, but not capacity- and demand-matched tests against direct scenario-risk controls.
This study	Frozen single- and double-link service loss; expected, CVaR, and worst-case risk	Four separated topology roles, held-out traffic matrices and failure pairs, topology-level inference	Estimates the residual prescriptive value of curvature after paths, MLU, latency, and failure information are controlled.

simple paths $\mathcal{P}_k$ for every method. Variable $x_{kp} \geq 0$ denotes the amount of commodity k assigned to path p .

3.2. Minimum-MLU stage

Every policy first solves the same linear program:

$$\min_{x,U} U, \quad (2)$$

$$\text{s.t.} \quad \sum_{p \in \mathcal{P}_k} x_{kp} = d_k, \quad \forall k \in K, \quad (3)$$

$$\sum_{k \in K} \sum_{p \in \mathcal{P}_k: a \in p} x_{kp} \leq U c_a, \quad \forall a \in A, \quad (4)$$

$$x_{kp} \geq 0, \quad U \geq 0. \quad (5)$$

This produces U^* . Unlike an ex-post overflow proxy, capacity is part of the optimization and OD commodities cannot cancel at intermediate nodes.

3.3. Matched-budget structural regularization

For a nonnegative normalized arc score r_a , the second stage constrains $U \leq (1 + \epsilon)U^*$ with $\epsilon = 0.02$ and minimizes

$$\sum_{k,p} x_{kp} \left(\frac{L_p}{\widetilde{L}} + \lambda \frac{R_p}{\widetilde{R}} \right), \quad L_p = \sum_{a \in p} \ell_a, \quad R_p = \sum_{a \in p} r_a, \quad (6)$$

where tildes denote within-instance scale factors. We test:

- endpoint-degree product;
- edge betweenness;
- negative Forman curvature, $r_e = [-F_e]_+$;
- negative Ollivier curvature, $r_e = [-\kappa_e]_+$;
- a seeded random-placebo score.

Curvature is computed on the undirected physical projection and assigned to both directed arcs. Ollivier curvature is calculated exactly with $\alpha = 0.5$ by solving the local Earth Mover’s Distance problem. For each policy, $\lambda \in \{0.03, 0.1, 0.3, 1, 3, 10, 30\}$ is selected using training traffic matrix 0000 only. A setting is eligible if training mean path cost is no more than 5% above the minimum-MLU reference; among eligible settings, the one with minimum score exposure is selected.

3.4. Direct scenario-risk controls

The single-link control was designed after inspecting the first confirmatory secondary outcome and frozen before evaluation on a third, disjoint topology set. The submission upgrade generalizes it to an explicit design-scenario set S . Let $h_{skp} = 1$ if path p intersects one or more physical links in failure scenario F_s , and zero otherwise. Counting the path once prevents a two-link scenario from double-counting traffic whose path contains both links. Normalized frozen loss is

$$L_s(x) = \frac{1}{D} \sum_{k,p} h_{skp} x_{kp}, \quad D = \sum_k d_k. \quad (7)$$

Within the same MLU and mean-path-cost budgets, the second lexicographic stage optimizes one of

$$\sum_{s \in S} \pi_s L_s(x), \quad \text{expected loss,} \quad (8)$$

$$\eta + \frac{1}{1 - \beta} \sum_{s \in S} \pi_s \xi_s, \quad \xi_s \geq L_s(x) - \eta, \quad \xi_s \geq 0, \quad \text{CVaR}_\beta, \quad (9)$$

$$Z, \quad L_s(x) \leq Z \quad \forall s, \quad \text{minimax loss.} \quad (10)$$

We set $\beta = 0.90$ and use uniform π_s in the design set. A third stage minimizes path cost without worsening the attained risk. The original single-link tail control is exactly the minimax case in which each F_s contains one physical link. These objectives test whether any apparent benefit is unique to curvature or follows more directly from optimizing the outcome.

3.5. Optimization procedure, guarantees, and size

For each topology and traffic matrix, the scenario-risk procedure is:

1. construct one method-neutral candidate-path catalog and solve exact minimum MLU;
2. impose the common MLU allowance and the explicit mean-path-cost limit;
3. build the binary scenario-path matrix $H = [h_{skp}]$ from design scenarios only;
4. minimize expected loss, CVaR, or maximum loss over this common feasible set;

5. fix the attained risk value within numerical tolerance and minimize mean path cost; and
6. evaluate the installed routing only on held-out traffic matrices and failure scenarios. Adaptive residual delivery is solved separately and is never an optimization target of a frozen policy.

This construction gives two useful guarantees. First, single-link minimax is not a separate heuristic: it is the special case of Eq. (10) with singleton F_s and the maximum-loss functional. Second, let $\mathcal{F}$ be the common path-flow feasible set after fixing the MLU and latency allowances, and let R_S be any of the three declared design-set risk functionals. If $x_S^* = \arg \min_{x \in \mathcal{F}} R_S(x)$, then $R_S(x_S^*) \leq R_S(x)$ for every feasible curvature or structural policy x . This design-set dominance is immediate from optimality; it supplies no held-out guarantee, which is why the evaluation scenario set is disjoint.

Let P be the number of path-flow variables, A the number of directed arcs, K the number of commodities, and S the number of explicit scenarios. The expected-loss LP uses $P + 1$ variables (flows and MLU), minimax uses $P + 2$, and CVaR uses $P + S + 2$ (including η and S excess variables). The CVaR and minimax risk stages contain K flow-conservation equalities and at most $A + S + 2$ inequalities before bounds; fixing optimal risk for the latency tie-break adds one inequality. The scenario-path matrix has at most SP nonzeros. Thus the explicit formulation is polynomial in the supplied path-scenario representation; the empirical timing study reports where this representation remains tractable rather than claiming Internet-scale control.

4. Failure outcomes and experimental design

4.1. Independent outcomes

For failure set $F \subseteq \mathcal{E}_{\text{phys}}$, frozen-route delivered fraction is

$$\text{Delivery}_{\text{frozen}}(F) = \frac{1}{\sum_k d_k} \sum_k \sum_{p: p \cap F = \emptyset} x_{kp}. \quad (11)$$

The primary outcome is service loss, $1 - \text{Delivery}_{\text{frozen}}$, averaged over every physical single-link failure and all held-out matrices.

For adaptive delivery, failed links are removed from the common candidate-path catalog and a residual LP first maximizes capacity-feasible delivered traffic. Only after the maximum delivery is fixed is policy cost used as a tie-break. This lexicographic order prevents a method from receiving credit for renaming its own structural score as resilience.

4.2. REPETITA data and separation of roles

We use the REPETITA ‘2016TopologyZooUCL_inverseCapacity’ dataset at repository commit ‘60e679c’. Directed capacities, delays, and OD directions are preserved. Ten development topologies were used to debug the protocol and are excluded from confirmatory inference. Before viewing outcomes, a second set of ten strongly connected topologies (17–23 nodes) and the implementation hashes were frozen. Matrix 0000 tunes λ ; matrices 0001–0004 are held out.

The confirmatory topologies are Aconet, Easynet, Noel, Belnet2010, Garr200109, Fccn, Packetexchange, GtsRomania, Fatman, and Nextgen. Full inclusion rules, exclusions, hashes, and the analysis decision are retained with the artifact.

After the direct tail-risk control was designed, a third set of ten previously unused topologies was frozen for its primary comparison with ORC: Claranet, HiberniaUk, Nordu2010, Spirallight, Garr199901, Grena, Hostway-International, Marwan, Peer1, and Rhnet. No outcome on this set was viewed before its protocol and implementation hashes were recorded.

For multi-failure validation, a fourth, previously unused set of twelve topologies was frozen: Sago, Pacificwave, Ibm, Highwinds, Internetmci, Restena, Marnet, BtAsiaPac, Harnet, Bandcon, HiberniaUs, and Azrena. All unordered pairs of physical links form the scenario universe. A SHA-256 hash of topology and sorted link-pair labels assigns each scenario to a design or evaluation set; the 160 smallest hashes per set are retained when necessary. The assignment uses no curvature, traffic outcome, or routing solution. Matrices 0001–0004 and every evaluation scenario remain hidden from all scenario-risk optimizers.

Before running additional robustness checks, we fixed a separate sensitivity protocol. Holding the same evaluation pairs constant, it changes CVaR level to $\beta \in \{0.80, 0.95\}$ at 160 design scenarios and changes the hash-ordered design budget to 40 or 80 scenarios at $\beta = 0.90$. These checks are labeled post-upgrade sensitivity rather than new confirmatory tests. Adaptive recovery is computed on the first 40 hash-ordered evaluation pairs per topology and all four held-out matrices; this cost-limited sample is identical for every score and frozen policy comparison.

Table 2 summarizes the confirmatory instances and the number of independent single-link failure cases used for each topology.

Table 2: Held-out confirmatory REPETITA instances. Failure cases equal four test traffic matrices times all physical single-link failures. Parallel bundles contain only equal-delay, equal-IGP-weight arcs.

Topology	Nodes	Directed arcs	Physical links	Bundles	OD pairs	Failure cases	λ_{ORC}
Aconet	23	62	31	0	506	124	1
Easynet	19	52	26	6	342	104	0.3
Noel	19	50	25	0	342	100	1
Belnet2010	22	50	25	10	462	100	0.3
Garr200109	22	48	24	0	462	96	0.03
Fccn	23	50	25	4	506	100	1
Packetexchange	21	54	27	0	420	108	0.3
GtsRomania	21	48	24	0	420	96	0.3
Fatman	17	42	21	0	272	84	1
Nextgen	17	38	19	2	272	76	1

4.3. Inference

The experimental unit is the topology. For topology g , the primary paired effect is

$$\Delta_g = L_{\text{ORC},g} - L_{\text{MLU},g}, \quad (12)$$

where L is mean frozen-route service loss across all held-out matrices and physical single-link failures. A negative value favors ORC. The frozen primary test is an exact two-sided sign-flip permutation test at $\alpha = 0.05$. We report the equal-topology mean and a topology-cluster bootstrap confidence interval [29]. Links are not treated as independent observations. Secondary policy-vs-MLU tests use Holm correction [30].

The frozen multi-failure primary endpoint is topology-level CVaR(0.90) across the Cartesian product of four held-out traffic matrices and held-out double-link failures. Its paired contrast is double-failure CVaR TE minus ORC; negative values favor the direct risk method. We again report an equal-topology mean, median, topology-cluster bootstrap interval, exact two-sided sign-flip test, and better/worse/tied counts. Mean and maximum loss, comparisons with MLU, and adaptive recovery are secondary; Holm correction is applied within each secondary outcome family.

5. Results

5.1. Confirmatory average-failure result

Table 3 reports the held-out topology-clustered effects. The ORC point estimate is in the harmful direction: mean failure loss is 0.452 percentage

Table 3: Confirmatory frozen-route service-loss effects relative to minimum MLU. Positive values are worse. Confidence intervals resample topologies. Holm p applies to the secondary five-policy family; the preregistered ORC primary test uses the raw exact $p = 0.0625$.

Policy	Mean Δ (pp)	95% CI (pp)	Path cost	Holm p
Degree	+0.270	[-0.094, +0.782]	+1.17%	0.8438
Betweenness	+0.205	[-0.159, +0.717]	+1.30%	0.8438
Forman–Ricci	+0.397	[+0.123, +0.707]	+2.28%	0.2344
Ollivier–Ricci	+0.452	[+0.077, +0.935]	+1.36%	0.2500
Random placebo	+0.197	[-0.089, +0.518]	+2.70%	0.8438

points higher than minimum MLU, and mean path cost is 1.36% higher. The exact primary p -value is 0.0625; therefore, under the frozen decision rule, we do not reject the null. This is evidence of no demonstrated improvement, not an equivalence conclusion. The percentile bootstrap interval and exact sign-flip test differ at the 5% boundary because only ten topology clusters are available and the paired effects are skewed; the decision follows the frozen exact test rather than selecting the more favorable inferential summary.

No secondary policy survives family-wise correction. Importantly, neither curvature score improves mean service loss. A structural penalty can change where traffic is carried while leaving the independent delivery outcome unchanged or worse. Figure 1 displays the topology-level heterogeneity behind this mean effect and the frozen exact-test result.

5.2. Held-out worst-failure validation

The first confirmatory set suggested a heterogeneous worst-failure signal, so the direct tail-robust control was designed and frozen for a new held-out test. On that third set, tail-robust reduced worst-failure loss relative to ORC by 2.246 percentage points (bootstrap 95% CI: -3.596 to -0.871; exact two-sided sign-flip $p = 0.03125$). It was better on six, worse on two, and tied on two topologies. Mean failure-loss difference was -0.103 points ($p = 0.5078$), and path-cost difference was -0.433% ($p = 0.6016$); neither was detected as non-zero.

Relative to minimum MLU, tail-robust reduced worst loss by 2.543 points (CI: -3.661 to -1.333; $p = 0.0078$), while mean loss was essentially unchanged (+0.012 points; $p = 0.7891$) and path cost rose by 1.70%. The result resolves

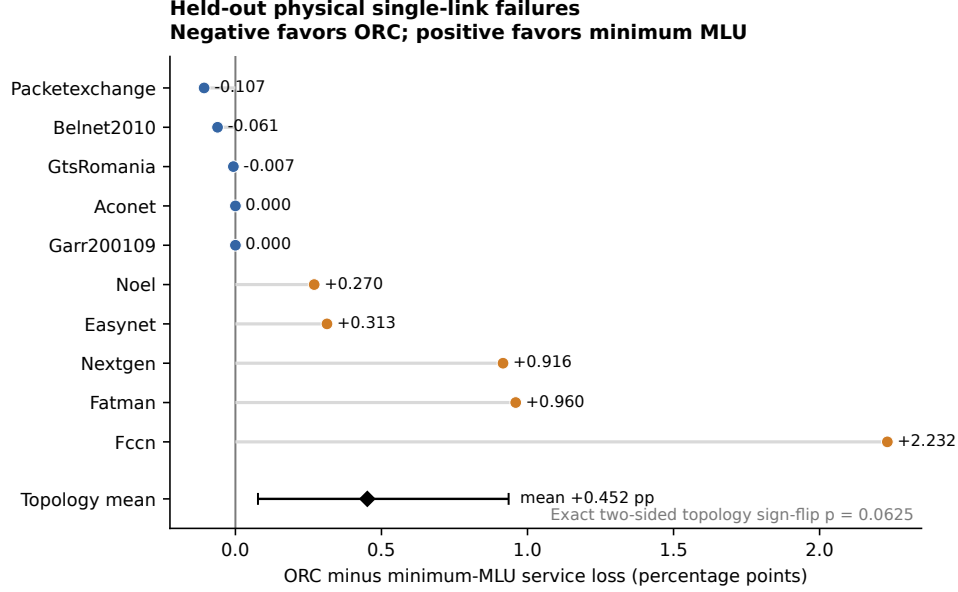


Figure 1: Topology-level confirmatory effects for Ollivier–Ricci regularization relative to minimum MLU. Positive values indicate greater frozen-route service loss under exhaustive physical single-link failures. The diamond and interval show the equal-topology mean and topology-cluster bootstrap 95% confidence interval.

the mechanism on held-out data: the tail benefit is not unique to curvature. A linear objective that represents frozen-route failure exposure directly is more effective. Figure 2 shows the topology-wise trajectories and their equal-topology means under the three policies.

5.3. Held-out double-failure scenario risk

The scenario-risk extension was evaluated on twelve further topologies. Design and evaluation link pairs were disjoint, and the evaluation pairs were never provided to an optimizer. The frozen primary contrast supported the direct risk control: relative to ORC, double-failure CVaR TE reduced held-out CVaR(0.90) service loss by 1.297 percentage points (topology-cluster bootstrap 95% CI: -2.074 to -0.554; exact two-sided sign-flip $p = 0.015625$). It was better on seven, worse on three, and tied on two topologies. Figure 3 shows that this mean is supported by seven negative topology-level effects rather than by one extreme instance.

This tail improvement is not a free average-loss improvement. Relative to

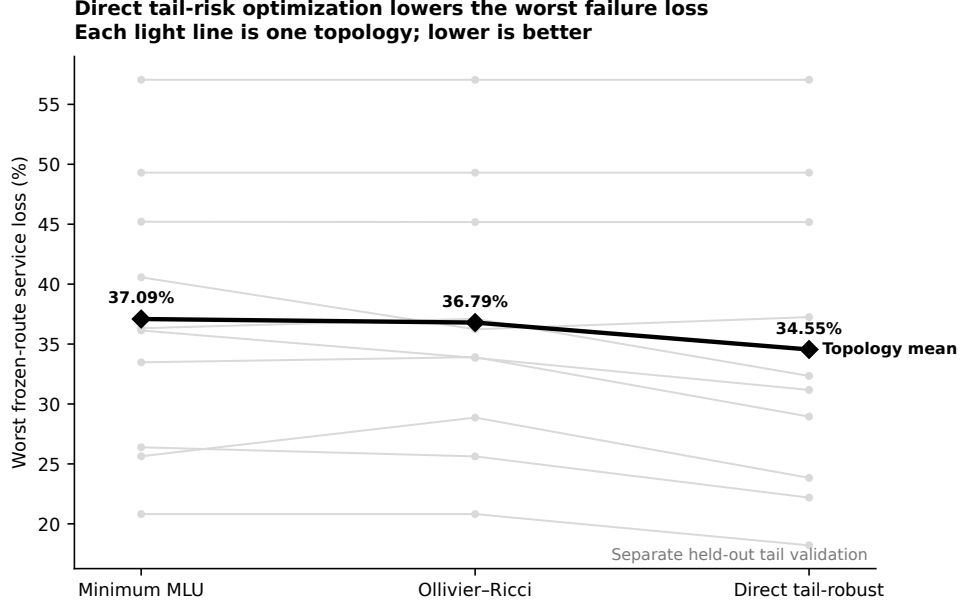


Figure 2: Held-out validation of worst-failure service loss on a third topology set. Each light line represents one topology under the three policies; the dark line is the equal-topology mean. The direct tail-robust LP reduces worst loss more consistently than the curvature regularizer.

ORC, the CVaR policy’s mean loss was 0.281 points higher (95% CI: +0.044 to +0.543; raw $p = 0.0664$) and mean path cost was 1.90% higher. Relative to minimum MLU, its CVaR improvement was 0.881 points (CI: -1.474 to -0.283; raw $p = 0.0215$, Holm-adjusted $p = 0.0645$), while mean loss increased by 0.648 points (Holm-adjusted $p = 0.0234$) and path cost by 3.76%. The formulation therefore does what its objective declares—moves mass out of the upper loss tail—but does not dominate on the mean.

Table 4 also shows that the simpler single-link minimax control generalizes strongly to unseen double failures. It reduced CVaR relative to ORC by 1.353 points (Holm-adjusted $p = 0.0156$) and worst loss by 2.319 points, with slightly lower path cost. Expected-loss optimization did not reduce the tail. The comparative mechanism is consequently broader than one new LP: direct failure exposure is a more useful prescriptive signal than curvature in these benchmarks, and the chosen risk functional determines which part of the loss distribution improves.

Table 4: Held-out double-link-failure effects relative to ORC over twelve new topologies. Negative values favor the candidate. CVaR intervals resample topologies. The double-CVaR comparison is the frozen primary test and uses its raw exact p ; other adjusted values use Holm correction within the CVaR secondary family.

Candidate	CVaR Δ (pp)	95% CI (pp)	Mean Δ (pp)	Path cost	p
Single-link minimax	-1.353	[-2.172, -0.635]	-0.272	-0.84%	0.0156
Double expected loss	-0.261	[-0.839, +0.301]	-0.515	-0.21%	0.4082
Double CVaR(0.90)	-1.297	[-2.074, -0.554]	+0.281	+1.90%	0.0156*
Double minimax	-1.009	[-1.838, -0.223]	-0.126	+0.53%	0.0938

*Frozen primary raw exact p ; no selection across policies.

Table 5: Frozen one-factor sensitivity of ORC minus minimum-MLU mean single-link service loss. Positive values are worse for ORC.

Condition	Mean Δ (pp)	95% CI (pp)	Path cost
$q = 4$	+0.522	[+0.155, +0.942]	+1.83%
$q = 16$	+0.452	[+0.079, +0.941]	+1.56%
MLU slack 0%	+0.442	[+0.081, +0.894]	+1.32%
MLU slack 5%	+0.465	[+0.075, +0.968]	+1.43%
ORC $\alpha = 0.25$	+0.452	[+0.080, +0.930]	+1.36%
ORC $\alpha = 0.75$	+0.452	[+0.082, +0.937]	+1.36%

5.4. Sensitivity, feasibility, and runtime

The confirmatory ORC-versus-MLU conclusion is insensitive to every frozen one-factor change (Table 5). ORC’s topology-mean single-failure loss remains higher under $q = 4$ and $q = 16$ paths, zero and 5% MLU slack, and $\alpha = 0.25$ and 0.75. Effect magnitudes range from +0.442 to +0.522 percentage points; no setting reverses the conclusion or removes its path-cost increase.

The constructive double-failure result is also stable to CVaR level and to a moderate reduction in design scenarios (Table 6). At 160 design scenarios, both $\beta = 0.80$ and 0.95 retain a lower held-out CVaR(0.90) than ORC. Reducing the design budget to 80 retains a -1.100-point effect; at 40 scenarios the estimate remains negative (-0.310 points) but its interval crosses zero. The attenuation is operationally informative: direct risk control needs adequate scenario coverage and is not a free consequence of the LP form. Every condition also retains a small mean-loss and path-cost penalty, so none establishes distribution-wide dominance.

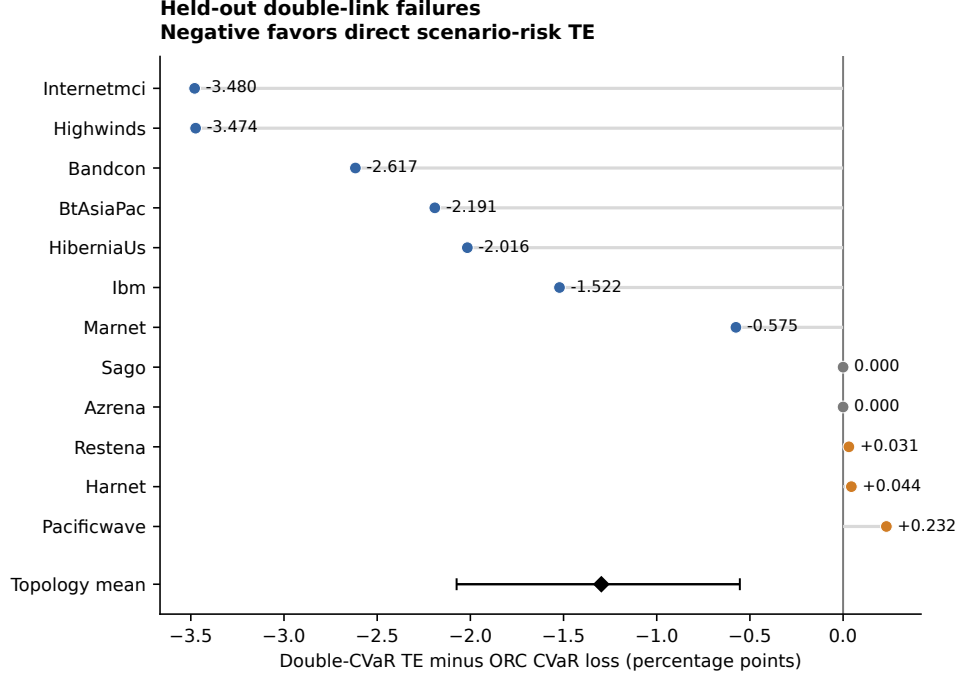


Figure 3: Topology-level primary effects for double-failure CVaR TE relative to ORC on held-out link-pair scenarios. Negative values indicate lower CVaR(0.90) service loss under the direct scenario-risk policy.

Adaptive recovery materially lowers loss but does not erase the frozen-routing comparison (Table 7). The capacity-feasible ceiling reduces equal-topology mean loss from about 20.3–20.9 points under the stored policies to 12.210 points, leaving an 8.117–8.713-point recoverable gap. On these same adaptive scenarios, topology-level mean Spearman association with residual loss is +0.428 for Ollivier risk (bootstrap 95% CI: +0.316 to +0.548), +0.305 for Forman risk, +0.182 for degree risk, and +0.551 for betweenness. Betweenness is therefore the strongest of the four descriptors; the adaptive association is structural but not uniquely geometric.

The twelve-topology double-failure run required 1,384 s wall time on a single-process Windows workstation with an Intel Core i9-13900H and 15.7 GB RAM. Across topologies, CVaR solves accounted for 155 s, adaptive recovery for 324 s, and exhaustive held-out evaluation for 432 s. Python allocator peaks do not include native HiGHS memory and are therefore reported

Table 6: One-factor sensitivity of held-out double-failure CVaR TE relative to ORC. Negative CVaR differences favor direct risk control. The main condition is confirmatory; the other rows are protocol-fixed robustness checks.

Condition	β	Design scenarios	CVaR(0.90) Δ (pp)	95% CI (pp)	Mean Δ / path cost
Main	0.90	up to 160	-1.297	[-2.074, -0.554]	+0.281 / +1.90%
CVaR level	0.80	up to 160	-1.343	[-2.193, -0.490]	+0.126 / +1.59%
CVaR level	0.95	up to 160	-1.212	[-2.056, -0.393]	+0.253 / +1.94%
Scenario budget	0.90	up to 40	-0.310	[-1.169, +0.528]	+0.278 / +1.16%
Scenario budget	0.90	up to 80	-1.100	[-1.827, -0.398]	+0.299 / +1.72%

Table 7: Frozen routing and the capacity-feasible adaptive recovery ceiling on the same 40 held-out double-failure pairs per topology and four test matrices. Values are equal-topology means. Recoverable gap is frozen mean loss minus adaptive mean loss.

Routing state	Mean loss (pp)	CVaR(0.90) (pp)	Maximum loss (pp)	Recoverable gap (pp)
Adaptive capacity-feasible ceiling	12.210	30.794	39.517	—
Frozen minimum MLU	20.328	38.245	45.386	8.117
Frozen Ollivier-Ricci	20.332	38.785	45.747	8.121
Frozen single-link minimax	20.478	37.593	44.070	8.267
Frozen double-failure CVaR	20.923	37.536	44.883	8.713

only as lower bounds.

The fixed $q = 4$ feasibility sequence (Table 8) shows that path variables and commodity count are more informative than node count alone. Four instances through 69 nodes solved successfully; Surfnets was the slowest at 89.4 s for 9,720 path variables. Garr201112 was retained in the fixed sequence but rejected by the lossless parser because it contains unequal-cost parallel arcs. We report this boundary rather than collapsing the arcs or substituting a more favorable topology.

6. Discussion

6.1. Structural diagnosis is not routing utility

Negative curvature often highlights edges between dissimilar neighborhoods. That may be useful for visualization, topology auditing, or generating failure hypotheses. Routing, however, is constrained by directed capacity, the traffic matrix, alternate paths, and the operational objective. A structural score that correlates with failure impact need not produce a better flow allocation when inserted as a penalty.

The controlled baselines expose this gap. Curvature regularization consumes latency budget and changes path allocation, yet it does not improve

Table 8: Fixed scalability sequence, matrix 0000 and $q = 4$. Total includes parsing, paths, ORC, tuning, and four final policies. Python peak excludes native solver allocations.

Topology	Nodes	Commodities	Path vars.	Paths (s)	CVaR (s)	Total (s)	Python peak
GtsHungary	30	867	1,794	0.81	4.91	6.98	21.2 MB
Geant2012	40	1,556	6,153	12.82	16.74	52.60	46.4 MB
Surfnet	50	2,450	9,720	22.54	32.28	89.40	83.1 MB
Garr201112	61	—	—	—	—	—	unsupported arcs
Latnet	69	4,683	11,026	19.51	10.50	47.86	80.5 MB

the average held-out delivery outcome. The direct tail objective also explains why reduced flow on a selected structural class is insufficient evidence: if worst-link exposure is the actual concern, it can be represented and optimized without a geometric surrogate. The held-out double-failure study extends this conclusion: single-link exposure control transfers to compound failures, while explicit CVaR control shifts the upper tail at an observable mean-loss cost.

6.2. Implications for curvature-aware networking

Our results do not imply that discrete curvature has no networking value. They support narrower uses and stronger validation requirements:

- Curvature may serve as a low-information structural feature when traffic and capacity data are unavailable.
- Predictive claims should compare curvature with degree, betweenness, and placebo scores on held-out failure outcomes.
- Prescriptive claims should compare curvature-aware routing with objectives that directly optimize mean loss, worst loss, or a declared risk measure.
- Risk-aware claims should report the entire mean–tail–path–cost trade-off; improving CVaR is not evidence of first-order stochastic dominance.
- Any computational shortcut from Forman to Ollivier curvature should report screening recall and should not describe uncomputed edges as exact ORC.

6.3. Limitations

The experiments use candidate-path multi-commodity flow rather than packet-level queues or a hardware testbed. Frozen delivery models the interval before route replacement; adaptive delivery models an ideal capacity-feasible controller and does not include detection or update delay. REPETITA traffic matrices and capacities are benchmark data rather than live operator telemetry. The first confirmatory and tail-validation sets each contain ten topologies, while the double-failure set contains twelve; these cluster counts still limit power for heterogeneity analysis. The tail-risk control was designed adaptively, although its reported effects were subsequently tested on disjoint frozen topology and scenario sets. Uniform design-scenario probabilities are a stress model, not estimates of operator-specific joint failure probabilities. The scenario-budget sensitivity shows that 40 design pairs are insufficient for a precise held-out tail gain, and the largest supported fixed timing instance has 69 nodes; the evidence therefore supports tractability on these backbone benchmarks, not Internet-scale or real-time deployment.

These limitations bound the claims. We report delivered traffic and utilization, not packet loss, jitter, controller convergence, or “self-healing.”

7. Conclusion

This study asks a prescriptive question that structural network geometry alone cannot answer: does curvature improve routing under actual link failures? In a matched-budget, held-out REPETITA experiment, Ollivier–Ricci regularization did not improve average single-link-failure service delivery beyond minimum-MLU TE. An apparent worst-failure benefit was exceeded by a direct tail-robust linear program at comparable path cost. On a fourth set with held-out double-link scenarios, direct CVaR optimization reduced upper-tail loss relative to ORC, although it increased mean loss, and the single-link minimax control also generalized. The tail effect persisted across CVaR levels but weakened under a 40-scenario design budget; ideal adaptive recovery left a substantial gap between stored and residual-capacity routing. The central lesson is methodological and operational: curvature can describe a network, but resilience must be evaluated with independent failures, and routing claims must survive comparison with objectives that directly represent the desired outcome and its trade-offs.

Data and code availability

The implementation, unit tests, frozen protocols, change log, exact JSON outputs, and scripts used to generate every table and figure are assembled as a versioned reproducibility artifact. The development repository is available at <https://github.com/emreozi/failure-aware-ricci-te>, and the exact v3.0.0 submission artifact is archived on Zenodo at [doi:10.5281/zenodo.21375484](https://doi.org/10.5281/zenodo.21375484). The external REPETITA source commit and dataset path are recorded in the artifact; benchmark data retain their original attribution and license [3].

CRedit authorship contribution statement

Emre Öztürk: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review and editing.

Declaration of competing interest

The author declares no competing financial or personal interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the author used Anthropic Claude and OpenAI Codex for language editing, code-review assistance, manuscript organization, and preparation of reproducibility materials. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

- [1] B. Fortz, M. Thorup, Internet traffic engineering by optimizing ospf weights, in: Proceedings of the 19th IEEE Conference on Computer Communications (INFOCOM), 2000, pp. 519–528. doi:10.1109/INFCOM.2000.832225.

- [2] H. H. Liu, S. Kandula, R. Mahajan, M. Zhang, D. Gelernter, Traffic engineering with forward fault correction, in: Proceedings of the 2014 ACM Conference on SIGCOMM, 2014, pp. 527–538. doi:10.1145/2619239.2626314.
- [3] S. Gay, P. Schaus, S. Vissicchio, REPETITA: Repeatable experiments for performance evaluation of traffic-engineering algorithms (2017). arXiv:1710.08665, doi:10.48550/arXiv.1710.08665.
- [4] D. Applegate, E. Cohen, Making intra-domain routing robust to changing and uncertain traffic demands: Understanding fundamental trade-offs, in: Proceedings of the 2003 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications, 2003, pp. 313–324. doi:10.1145/863955.863991.
- [5] H. Wang, H. Xie, L. Qiu, Y. R. Yang, Y. Zhang, A. Greenberg, COPE: Traffic engineering in dynamic networks, in: Proceedings of the 2006 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications, 2006, pp. 99–110. doi:10.1145/1151659.1159926.
- [6] Y. Wang, H. Wang, A. Mahimkar, R. Alimi, Y. Zhang, L. Qiu, Y. R. Yang, R3: Resilient routing reconfiguration, in: Proceedings of the ACM SIGCOMM 2010 Conference, 2010, pp. 291–302. doi:10.1145/1851182.1851218.
- [7] J. Bogle, N. Bhatia, M. Ghobadi, I. Menache, N. Bjørner, A. Valadarsky, M. Schapira, TeaVaR: Striking the right utilization–availability balance in WAN traffic engineering, in: Proceedings of the ACM Special Interest Group on Data Communication, 2019, pp. 29–43. doi:10.1145/3341302.3342069.
- [8] Y. Perry, F. V. Frujeri, C. Hoch, S. Kandula, I. Menache, M. Schapira, A. Tamar, DOTE: Rethinking (predictive) WAN traffic engineering, in: 20th USENIX Symposium on Networked Systems Design and Implementation, USENIX Association, 2023, pp. 1557–1581.
URL <https://www.usenix.org/conference/nsdi23/presentation/perry>

- [9] P. Namyar, A. Ghavidel, D. Crankshaw, D. S. Berger, K. Hsieh, S. Kandula, R. Govindan, B. Arzani, Enhancing network failure mitigation with performance-aware ranking, in: 22nd USENIX Symposium on Networked Systems Design and Implementation, USENIX Association, 2025, pp. 335–357.
URL <https://www.usenix.org/conference/nsdi25/presentation/namyar>
- [10] T. Schneider, S. Vissicchio, L. Vanbever, Verifying maximum link loads in a changing world, in: 22nd USENIX Symposium on Networked Systems Design and Implementation, USENIX Association, 2025, pp. 1269–1287. doi:10.3929/ethz-b-000711664.
URL <https://www.usenix.org/conference/nsdi25/presentation/schneider>
- [11] Y. Ollivier, Ricci curvature of markov chains on metric spaces, *Journal of Functional Analysis* 256 (3) (2009) 810–864. doi:10.1016/j.jfa.2008.11.001.
- [12] R. Forman, Bochner’s method for cell complexes and combinatorial ricci curvature, *Discrete & Computational Geometry* 29 (3) (2003) 323–374. doi:10.1007/s00454-002-0743-x.
- [13] A. Samal, R. P. Sreejith, J. Gu, S. Liu, E. Saucan, J. Jost, Comparative analysis of two discretizations of ricci curvature for complex networks, *Scientific Reports* 8 (2018) 8650. doi:10.1038/s41598-018-27001-3.
- [14] C.-C. Ni, Y.-Y. Lin, J. Gao, X. D. Gu, E. Saucan, Ricci curvature of the internet topology, in: 2015 IEEE Conference on Computer Communications (INFOCOM), 2015, pp. 2758–2766. doi:10.1109/INFOCOM.2015.7218668.
- [15] S. Jain, A. Kumar, S. Mandal, J. Ong, L. Poutievski, A. Singh, S. Venkata, J. Wanderer, J. Zhou, M. Zhu, J. Zolla, U. Hölzle, S. Stuart, A. Vahdat, B4: Experience with a globally-deployed software defined WAN, in: *Proceedings of the ACM SIGCOMM 2013 Conference*, 2013, pp. 3–14. doi:10.1145/2486001.2486019.
- [16] C.-Y. Hong, S. Kandula, R. Mahajan, M. Zhang, V. Gill, M. Nanduri, R. Wattenhofer, Achieving high utilization with software-driven WAN,

- in: Proceedings of the ACM SIGCOMM 2013 Conference, 2013, pp. 15–26. doi:10.1145/2534169.2486012.
- [17] S. Gay, R. Hartert, S. Vissicchio, Expect the unexpected: Sub-second optimization for segment routing, in: IEEE INFOCOM 2017, 2017, pp. 1–9. doi:10.1109/INFOCOM.2017.8056971.
- [18] P. Kumar, Y. Yuan, C. Yu, N. Foster, R. Kleinberg, P. Lapukhov, C. L. Lim, R. Soulé, Semi-oblivious traffic engineering: The road not taken, in: 15th USENIX Symposium on Networked Systems Design and Implementation, 2018, pp. 157–170.
URL <https://www.usenix.org/conference/nsdi18/presentation/kumar>
- [19] Z. Zhong, M. Ghobadi, A. Khaddaj, J. Leach, Y. Xia, Y. Zhang, ARROW: Restoration-aware traffic engineering, in: Proceedings of the 2021 ACM SIGCOMM Conference, 2021, pp. 560–579. doi:10.1145/3452296.3472921.
- [20] U. Krishnaswamy, R. Singh, N. Bjørner, H. Raj, Decentralized cloud wide-area network traffic engineering with BLASTSHIELD, in: 19th USENIX Symposium on Networked Systems Design and Implementation, 2022, pp. 325–338.
URL <https://www.usenix.org/conference/nsdi22/presentation/krishnaswamy>
- [21] S. Knight, H. X. Nguyen, N. Falkner, R. Bowden, M. Roughan, The internet topology zoo, IEEE Journal on Selected Areas in Communications 29 (9) (2011) 1765–1775. doi:10.1109/JSAC.2011.111002.
- [22] S. Orlowski, R. Wessäly, M. Pióro, A. Tomaszewski, SNDlib 1.0—survivable network design library, Networks 55 (3) (2010) 276–286. doi:10.1002/net.20371.
- [23] Y. Chang, S. Rao, M. Tawarmalani, Robust validation of network designs under uncertain demands and failures, in: 14th USENIX Symposium on Networked Systems Design and Implementation, 2017, pp. 347–362.
URL <https://www.usenix.org/conference/nsdi17/technical-sessions/presentation/chang>

- [24] P. Namyar, B. Arzani, R. Beckett, S. Segarra, H. Raj, U. Krishnaswamy, R. Govindan, S. Kandula, Finding adversarial inputs for heuristics using multi-level optimization, in: 21st USENIX Symposium on Networked Systems Design and Implementation, USENIX Association, 2024, pp. 927–949.
URL <https://www.usenix.org/conference/nsdi24/presentation/namyar-finding>
- [25] R. T. Rockafellar, S. Uryasev, Optimization of conditional value-at-risk, *The Journal of Risk* 2 (3) (2000) 21–41. doi:10.21314/JOR.2000.038.
- [26] R. P. Sreejith, K. Mohanraj, J. Jost, E. Saucan, A. Samal, Forman curvature for complex networks, *Journal of Statistical Mechanics: Theory and Experiment* 2016 (6) (2016) 063206. doi:10.1088/1742-5468/2016/06/063206.
- [27] E. Saucan, R. P. Sreejith, R. P. Vivek-Ananth, J. Jost, A. Samal, Discrete ricci curvatures for directed networks, *Chaos, Solitons & Fractals* 118 (2019) 347–360. doi:10.1016/j.chaos.2018.11.031.
- [28] O. Chiriac, W. Harris, C. He, B. Foo, Hierarchical community detection and routing on networks with Forman–Ricci curvature and flow, in: 2026 IEEE Aerospace Conference, IEEE, 2026. doi:10.1109/AER066936.2026.11520050.
- [29] B. Efron, Bootstrap methods: Another look at the jackknife, *The Annals of Statistics* 7 (1) (1979) 1–26. doi:10.1214/aos/1176344552.
- [30] S. Holm, A simple sequentially rejective multiple test procedure, *Scandinavian Journal of Statistics* 6 (2) (1979) 65–70. doi:10.2307/4615733.